



SFWRENG 2XC3

SOFTWARE ENGINEER PRACTICE AND EXPERIENCE: DEVELOPMENT BASICS

Department of Computing and Software
Faculty of Engineering
McMaster University

Fall 2023

Welcome Section, SFWRENG 2XC3



- **Course Instructor**

Contact Information: yazdinea@mcmaster.ca

- **Abbas Yazdinejad**

- Designing the Course Content
 - Teaching: deliver lectures
 - Assessing Student Performance

- **Teaching Assistants (TAs)**

- Running Lab sessions
 - Office Hours
 - Address student Emails
 - Grading labs and final project

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Why SFWRENG 2XC3 is important?



- 1. Hands-On UNIX Experience:** UNIX-like systems are widely used in professional software development.
 - 2. Understanding of System Interactions:** The course covers pipes and filters, which are fundamental concepts in UNIX-like systems for data processing and system interaction.
 - 3. Foundation in Low-Level Imperative Programming:** This course provides a deep understanding of the fundamental concepts of low-level imperative programming.
 - 4. Scripting concept:** Shell programming allows for automation of tasks, batch processing, and system management.
 - 5. C Programming:** C remains one of the most widely used programming languages, C provides a strong foundation for understanding how software interacts with hardware.
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Who Should Attend?

1.Aspiring Software Engineers: Anyone looking to build a career in software development will benefit from the foundational knowledge this course provides.

2.Computer Science Students: Those who have a theoretical understanding of computing but want to gain practical experience in software development.

3.System Administrators: Those who manage and maintain computer systems will find the UNIX interaction and shell programming skills invaluable.

4.Embedded Systems Developers: With a focus on low-level programming in C, this course is perfect for those aiming to work in embedded systems or hardware interfacing.

5.Students Seeking Advanced Programming Skills



Which contents will we cover in the course?

Topics include:

- UNIX interaction, including remote access
- Shell programming (bash shell)
- Low-level programming in C:
 - Number representation Stack/heap memory model
 - Arrays and pointers
 - Heap-based data structures (linked lists, trees)
 - Modularization conventions, separate compilation, make

Schedule



	Section	Day	Time slot	Location
Lectures	C01	Mon	9:30AM - 10:20AM	MDCL 1105
Labs	L01	Mo-We	2:30PM - 4:20PM	JHE 233A
	L02	Tu-Th	2:30PM - 4:20PM	JHE 233A
	L03	Tu-Th	8:30AM - 10:20AM	BSB 244

Schedule

- **First lecture** Sep 11, 2023
- **Last lecture** Dec 4, 2023

- **No lectures** during Midterm recess (reading week) Oct 9-15
- **No labs** until Sep 18
- **No labs** during Midterm recess (reading week) Oct 9-15

- **Practice labs** in the period Sep 18-19
- **Regular labs** in the period Sep 25- Nov 21
- **Makeup labs** in the period Nov 27-28
- **Final project** release: Nov 7 due Dec 4 (by 11 PM)

Detailed lab schedule by sections

Lab materials	Section L01	Section L02	Section L03
Practice A	Sep 18	Sep 19	Sep 19
Practice B	Sep 20	Sep 21	Sep 21
Minor 1A	Sep 25	Sep 26	Sep 26
Minor 1B	Sep 27	Sep 28	Sep 28
Minor 1C	Oct 2	Oct 3	Oct 3
Major 1	Oct 4	Oct 5	Oct 5
Minor 2A	Oct 16	Oct 17	Oct 17
Minor 2B	Oct 18	Oct 19	Oct 19
Minor 2C	Oct 23	Oct 24	Oct 24
Major 2	Oct 25	Oct 26	Oct 26
Minor 3A	Oct 30	Oct 31	Oct 31
Minor 3B	Nov 1	Nov 2	Nov 2
Minor 3C	Nov 6	Nov 7	Nov 7
Major 3	Nov 8	Nov 9	Nov 9
Minor 4A	Nov 13	Nov 14	Nov 14
Minor 4B	Nov 15	Nov 16	Nov 16
Minor 4C	Nov 20	Nov 21	Nov 21
Major 4	Nov 22	Nov 23	Nov 23
Makeup A	Nov 27	Nov 28	Nov 28
Makeup B	Nov 29	Nov 30	Nov 30

} Set-up
 } optional,
 but HIGHLY
 recommended
 ← Mandatory

learning Resources and Class Policy

- **The class notes (slides)**
- **Instruction and lab walks-through**
- **Other existing resources:**
 - Bash Reference Manual: <https://www.gnu.org/software/bash/manual/bash.html>
 - C tutorial for beginners: <https://www.freecodecamp.org/news/the-c-beginners-handbook/>
- **Online Platforms/Courses:**
 - Unix, Shell programing and C programing

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- Cooperation among students is strongly encouraged during the minor labs, however the major lab projects and the final project are to be individual effort.
- The series of minor labs are followed by a major lab.
- The minor labs is open on Mondays, so all students have the same access regardless of the lab section.
- Checks for plagiarism will be performed for Major labs and final project.
- For all email communication with the instructor and the TAs, the students must use their official McMaster email account, emails from other accounts will be ignored.
- The different lab sections work on different major labs.



Student Assessment

Student Assessment (Grading)

Evaluation is based on two factors: 4 major lab projects and a final project according to the formula given below. Every student will have 5 marks at the end of the term, no exception. How a missed academic work will be dealt with, see below

4 major labs @ 19 %	76 %
Final project	24 %
Total	100 %